

Practice exercise, Lecture 7

ØKA2015 Applied Data Science. Not graded. Do it before the next lecture.

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Reference: the Lecture 7 slides and `lecture07_descriptive.R`. You use the same steps on a new table: the California housing data from Assignment 1, read straight from the web:

```
library(tidyverse)
url <- paste0("https://raw.githubusercontent.com/touseefhameed/",
              "oka2015-applied-data-science/main/assignment1/data/housing.csv")
housing <- read_csv(url)
```

`paste0()` glues the two pieces of text into one web address; it is split only so that it fits on the page. From your Assignment 1 project, `read_csv("data/housing.csv")` does the same.

Each row is a district in California. Work in a script, and write one sentence under each answer. Eight tasks, about 50 minutes.

Task 1: types of data

- Use `glimpse()`. Which columns are quantitative and which are categorical?
- Is `housing_median_age` discrete or continuous? Is `ocean_proximity` nominal or ordinal?
- Is this cross-sectional or time series data?

Task 2: a frequency table

Make a frequency table of `ocean_proximity` with a percent column, largest first. Which category is smallest, and why might that matter later?

Task 3: shape

Draw a histogram of `median_income` and one of `housing_median_age`. Describe each shape. Check your answer by comparing the mean and the median.

Task 4: centre

- The mean, median and mode of `median_house_value`.
- The same three measures for each `ocean_proximity` group, with `n`. Which group has the highest median value?

Task 5: spread

- The standard deviation, the range and the IQR of `median_house_value`.
- Use `summary()` on the same column. What does the maximum tell you?

Task 6: missing values

- Which column has missing values, and how many?
- Look at its shape. Would you fill the missing values with the mean or the median? Fill them, and check that none are left.

Task 7: outliers

- Draw a boxplot of `median_house_value`. Count the districts outside the IQR fences.
- Count the districts with a value of exactly 500,001. Is that a real price or a limit set by whoever collected the data? What would you do with them?
- Use z-scores to count the districts whose `total_rooms` is more than 3 sd from the mean.

Task 8: association

- Draw a scatter plot of `median_income` against `median_house_value`.
- The correlation between them. Strong or weak? Positive or negative?
- The correlation matrix of `median_income`, `median_house_value`, `housing_median_age` and `population`. Which pair is the weakest?
- Does a high correlation between income and house value prove that income causes high prices? Why not?

Answers to “Check yourself” from the script

- `ocean_proximity` is categorical. Its categories have no natural order, so it is nominal.
- Mean above median: skewed right. A few very expensive houses pull the mean up.
- The standard deviation is in the units of the data; the variance is in squared units.
- $IQR = 30 - 10 = 20$. Fences: $10 - 1.5 \times 20 = -20$ and $30 + 1.5 \times 20 = 60$.
- A clear downward cloud of points: as one variable rises, the other falls.

The answers to Tasks 1 to 8 are not distributed now. Compare with a classmate, or bring your script to the next lecture.